

Voltage-driven gated van der Pauw method for accurate channel and contact resistance extraction in oxide Thin-Film Transistors

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ABSTRACT

Precise characterization of both channel and contact characteristics in p- and n-type oxide thin-film transistors (TFTs) is essential for the development of next-generation low-power complementary metal–oxide–semiconductor circuits. However, p-type oxide TFTs remain limited by inherently low hole concentrations and inefficient charge injections at the contacts, leading to high contact resistance and performance limitation. To address these challenges, a voltage-driven gated van der Pauw (V-gVDP) method is proposed, enabling the simultaneous and accurate extraction of the channel sheet conductance (σ_s) and the specific contact resistivity (ρ_c) compared to the conventional current-driven gVDP or the transmission line method (TLM). Voltage differences in the V-gVDP enable spatial decoupling and independent evaluation of channel and contact characteristics using a single device, without the need for control devices, with superior precision and reproducibility. The relative standard deviation (RSD) of the extracted sheet resistance (R_s) from the V-gVDP was reduced by factors of 6 and 46 for SnO and IGZO, respectively, while the RSD of ρ_c was improved by factors of 51 and 10 compared to the TLM. The V-gVDP method was successfully applied to both p-type SnO and n-type IGZO TFTs, confirming its robustness and versatility across different carrier polarities. These results demonstrate the potential of the V-gVDP method as a reliable and high-precision platform for characterizing oxide semiconductors and optimizing CMOS circuits and contact engineering.

1. Introduction

Thin-film transistor (TFT) based complementary metal–oxide–semiconductor (CMOS) technology has emerged as a key technology for advanced electronic applications with low power consumption [1–4]. In particular, oxide semiconductors, with their low-temperature processability, high transparency, and low leakage current, enable the development of low-power and high-performance electronic devices, making them highly suitable for next-generation CMOS circuit development [5–12]. To realize such CMOS circuits, it is essential to develop p-type oxide TFTs such as tin oxide (SnO_x), copper oxide (CuO_x), and tellurium oxide (TeO_x) [13–18], in addition to the well-established n-type oxide TFTs represented by indium–gallium–zinc–oxide (IGZO). However, p-type oxide TFTs typically suffer from low hole mobility, high contact resistance (R_c), and

nonlinear contact behavior. The precise extraction of key parameters such as sheet conductance (σ_s), specific contact resistivity (ρ_c), and field-effect mobility is challenging when scaled down, which are essential for circuit-level modeling and performance optimization [19–23]. Although the conventional transmission line method (TLM) is used to extract an R_c value based on multiple cross-checks across various devices, this approach often results in incorrect channel parameter extraction due to device-to-device variations, and it becomes typically problematic when the channel length is reduced, leading to mobility overestimation [24–28]. This issue is particularly pronounced in p-type oxide TFTs, which tend to have a high R_c , making the accurate extraction of channel and contact properties a critical challenge for advancing the technology [8,13,22,23].

The current-driven gated van der Pauw (C-gVDP) method was introduced to overcome the limitations of the TLM analysis, allowing the

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extraction of intrinsic channel sheet resistance (R_S) independent of R_C , thereby providing a more accurate characterization of the channel [28–32]. However, the C-gVDP method still requires multiple and repeated measurements driven at specific current conditions, as well as the use of additional control devices for precise R_C extraction, making its practical implementation challenging.

In this study, we propose a voltage-driven gVDP (V-gVDP) method to precisely determine the individual contributions of contact and channel characteristics, thereby overcoming the limitations of the C-gVDP approach. The proposed V-gVDP method improves both the measurement reliability and reproducibility of the electrical characterization (σ_S and ρ_C) of both p- and n-type oxide TFTs. Compared to the TLM method, the relative standard deviation (RSD) of R_S was 6 times lower in SnO and 46 times lower in IGZO, while the RSD of ρ_C was reduced by factors of 51 and 10, respectively. These results indicate that the V-gVDP method offers significantly higher precision than TLM. Notably, whereas TLM often produces negative contact resistance values due to device-to-device variations, the V-gVDP method consistently yields physically valid values across all measured conditions. These findings, validated through both simulation and experimental data, confirm the robustness and general applicability of the proposed approach. The V-gVDP method thus serves as a precise and reliable characterization technique for optimizing oxide-based CMOS circuit design and low-power, high-reliability electronic devices.

2. Results and discussion

2.1. Transmission line method

The TLM is a widely used technique for extracting R_C in TFTs. This

method involves fabricating multiple devices with varying channel lengths (L) and linearly plotting the width-normalized total resistance (wR_T) as a function of L . R_C can be evaluated by extrapolating the linear fit of total resistance to $L = 0$, using the device resistance referenced to gate-overdrive voltage ($V_G - V_T$) [33]. Furthermore, the ρ_C can be calculated using the expression $\rho_C = \frac{(R_C W)^2}{R_S}$ [23,34,35]. However, the linearity assumed in this method is often violated in practical devices. When the R_C at the contact/channel interface is significant, charge injection becomes strongly dependent on the gate voltage (V_G), resulting in V_T shifts among devices and a loss of linear behavior [25,36,37]. This issue is particularly pronounced in p-type oxide TFTs due to the larger effective mass of holes, which reduces the tunneling probability and hinders charge injection at the contact. In addition, p-type oxide semiconductors typically exhibit lower carrier mobility, inferior interface quality, and higher trap densities compared to their n-type counterparts, further complicating consistent electrical behavior across devices and undermining the accuracy of TLM analysis [13,38,39]. Fig. 1 presents examples of both p-type SnO and n-type IGZO TFTs. Fig. 1a shows the transfer characteristics of SnO TFTs with a fixed channel width of 25 μm and varying channel lengths of 10, 25, 50, and 75 μm . As illustrated in Fig. 1b, the wR_T versus L plots hardly converge to a single intersection point upon extrapolation at different V_G values. Notably, the extracted R_C appears negative at $L = 0$. For instance, negative R_C values of -0.14 and -0.85 $\text{k}\Omega\text{cm}$ were obtained at $V_G - V_T = -28$ V and -30 V, respectively. This deviation disrupts the linearity necessary for reliable extraction of R_S and leads to inconsistencies in the wR_T values across devices. These results highlight that applying the conventional TLM to oxide TFTs is highly sensitive to device conditions and particularly unreliable for p-type TFTs, where charge injection is unstable. In contrast,

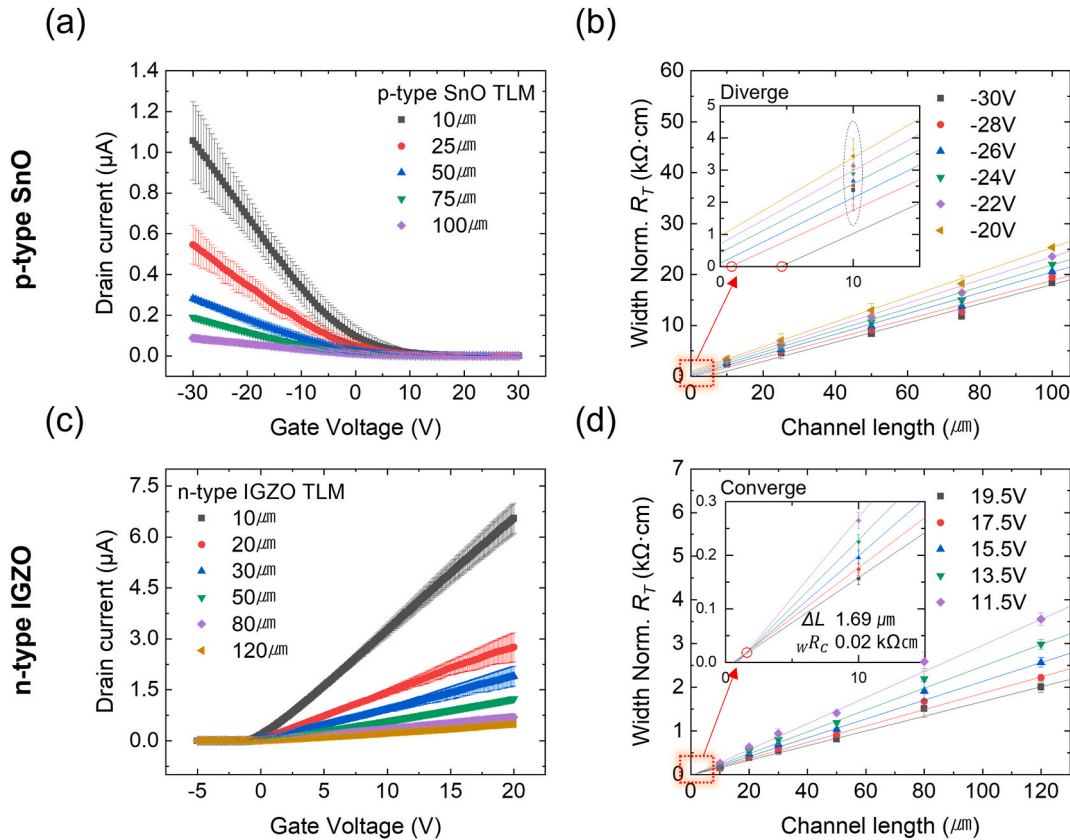


Fig. 1. Transmission line method (TLM) analysis of p-type SnO and n-type IGZO TFTs. (a) Transfer characteristics of SnO TFTs with a fixed channel width (W) of 25 μm and channel lengths (L) of 10, 25, 50, and 75 μm . (b) Width-normalized total resistance (wR_T) versus channel length (L) plots for SnO TFTs at various gate voltages (V_G). The data exhibits poor linearity and inconsistencies in extrapolation. (c) Transfer characteristics of n-type IGZO TFTs with a fixed W of 25 μm and L values of 10, 20, 30, 50, 80, 120 μm . (d) wR_T versus L fitting of IGZO TFTs at various V_G values, yielding $\Delta L = 1.69$ μm , $wR_C = 0.02$ $\text{k}\Omega\text{cm}$ and $\rho_C = 1 \times 10^{-3}$ Ωcm^2 .

Fig. 1c and d present the transfer characteristics and wR_T versus L plots of an n-type IGZO TFT measured under identical conditions. Unlike the p-type case, the IGZO TFTs exhibit more stable electrical behavior, allowing for linear extrapolation with a clearly defined common intersection point. From this analysis, the extracted parameters include $\Delta L = 1.69 \mu\text{m}$, wR_C of $0.02 \text{ k}\Omega \text{ cm}$, and ρ_C of $1 \times 10^{-3} \Omega \text{ cm}^2$. Importantly, accurate extraction of the device-specific V_T , which is critical for improving the reliability of the analysis, remains a significant challenge in the study [37,39,40].

2.2. Current-vs. voltage-driven gated van der Pauw methods

To overcome the limitations of TLM and enable more accurate analysis of R_C , a C-gVDP method was previously proposed [28]. This method allows for the precise extraction of sheet conductance ($\sigma_s = 1/R_s$) and intrinsic mobility (μ_{int}) by injecting a current source and measuring the voltage drop across a region of equal potential within the channel, referred to as the conductance regime. By decoupling the R_s measurement from device-to-device variability, the method enhances the accuracy of electrical characterization. Fig. 2a illustrates the conceptual schematic of the C-gVDP method. A current is applied to contact 1, while sweeping the V_G , and the voltage drop (V_{34}) between contacts 3 and 4, located in the equipotential conductance region indicated by the comb pattern, is measured to extract σ_s . In this configuration, contacts 3 and 4 function solely as voltage probes, sensing the potential distribution generated by the current (I_{12}) flowing between contacts 1 and 2. The measured potential corresponds to the equipotential lines at contacts 3 and 4 and reflects only the potential drop within the channel (V_{34}). As a result, the voltage drop at the electrode/channel interfaces, R_C , is excluded. As σ_s varies with the magnitude of the injected current, obtaining a continuous σ_s profile requires applying finely incremented current steps. As shown in Fig. S1, the current magnitude determines the measurable range of σ_s . Additionally, for each current level, eight configurations, such as I_{12} , I_{21} , I_{23} , I_{32} , I_{34} , I_{43} , I_{41} , and I_{14} , must be measured by rotating the current injection and sensing positions. Moreover, to analyze contact characteristics, the obtained σ_s must be applied to a separate device with identical channel characteristics, making the process time-consuming and highly sensitive to V_T shift and device degradation.

Instead, a V-gVDP method is proposed that employs a voltage source rather than a current source, enabling simultaneous and effectively decoupled measurement of channel and contact properties under a shared drain current within a single device. Fig. 2b shows the structure of the proposed V-gVDP method. In this configuration, a voltage is applied to contact 1 while sweeping V_G , thereby inducing a drain current between contacts 1 and 2, which defines the transfer regime. The voltage drops (V_{12}) measured in this regime reflect the R_T of the device, including both contact and channel contributions. Concurrently, the potential distribution established by the drain current flowing through the channel is measured as a voltage drop (V_{34}) between contacts 3 and 4, corresponding to the conductance regime, as in the C-gVDP method.

As noted above, this V_{34} represents only the internal channel voltage drop, excluding any contribution from R_C . Unlike the C-gVDP method, however, the drain current in the V-gVDP varies continuously with V_G , allowing the continuous behavior of V_{34} to be obtained without the need for stepwise current injections. In summary, V_{12} includes both R_{Ch} and R_C , while V_{34} isolates the channel resistance under equipotential conditions. Unlike C-gVDP, the V-gVDP method enables continuous extraction of the σ_s because V_{34} is measured as a function of the varying I_D during the V_G sweep. The transfer regime provides R_T from the I_D - V_G curve, while the conductance regime allows intrinsic R_s to be extracted from the σ_s vs. V_G - V_C curve, where V_C is the central channel potential defined as $(V_3 + V_4)/2$. By leveraging a shared drain current to probe two voltage-sensing regimes with distinct physical meanings, both channel and contact parameters can be independently extracted from a single device through a single V_G sweep, minimizing errors caused by device variability. Fig. 2c compares the conceptual schematics of the C-gVDP and V-gVDP methods. The C-gVDP method enables the extraction of σ_s from the conductance regime. However, determining R_C requires additional analysis using conventional TFTs fabricated in a different configuration from the gVDP structure. In contrast, the V-gVDP method integrates the principle of the C-gVDP to extract σ_s from the conductance regime and, by using a voltage source, additionally allows the direct extraction of R_T from the transfer regime using the same gVDP-configured device. This approach not only eliminates the need for fabricating separate TFT structures but also enables the simultaneous evaluation of σ_s and R_T from a single device, thereby facilitating a more accurate and reliable analysis. By combining these outputs, R_{Ch} , R_C , and ρ_C can all be determined independently and accurately. This voltage-driven method offers clear advantages in terms of practicality and precision, as it eliminates device-to-device variability and relies on direct measurements rather than trend-based extrapolation. As such, the V-gVDP method applies to both n-type and p-type oxide TFTs, serving as a universal and high-precision platform for characterizing and developing oxide-based CMOS circuits.

2.3. Voltage-driven gated van der Pauw method

Fig. 3a shows a top-view image of a V-gVDP device fabricated using p-type SnO. When a drain voltage (V_D) is applied to contact 1, current flows (I_D) from contact 1 to 2, simultaneously forming both the transfer and conductance regimes within the same device. Fig. 3b presents the conceptual layout of the conductance regime. The drain current (I_D) flows between contacts 1 and 2, while contacts 3 and 4 function solely as voltage probes within the channel. Accordingly, the measured voltage difference (V_{34}) excludes the influence of R_C and reflects only the potential distribution inside the channel. The corresponding resistance is calculated as $R_{34} = \frac{V_{34}}{I_{12}}$, which reflects only the intrinsic channel properties. According to van der Pauw's theory, the sheet conductance can then be determined as $\sigma_s = \frac{\ln 2}{\pi R_{34}}$. This measurement principle mirrors that of the C-gVDP method. The extracted σ_s values as a function of V_G -

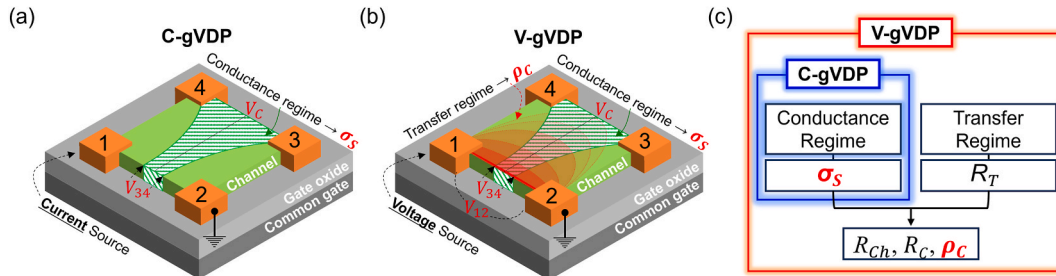


Fig. 2. 3D schematics of Si common-gate-based gVDP devices. (a) current-driven gVDP (C-gVDP) and (b) voltage-driven gVDP (V-gVDP) devices under operation. (c) Schematic comparison of the C-gVDP and V-gVDP methods, illustrating the differences in measurement configuration and the extraction of channel and contact parameters.

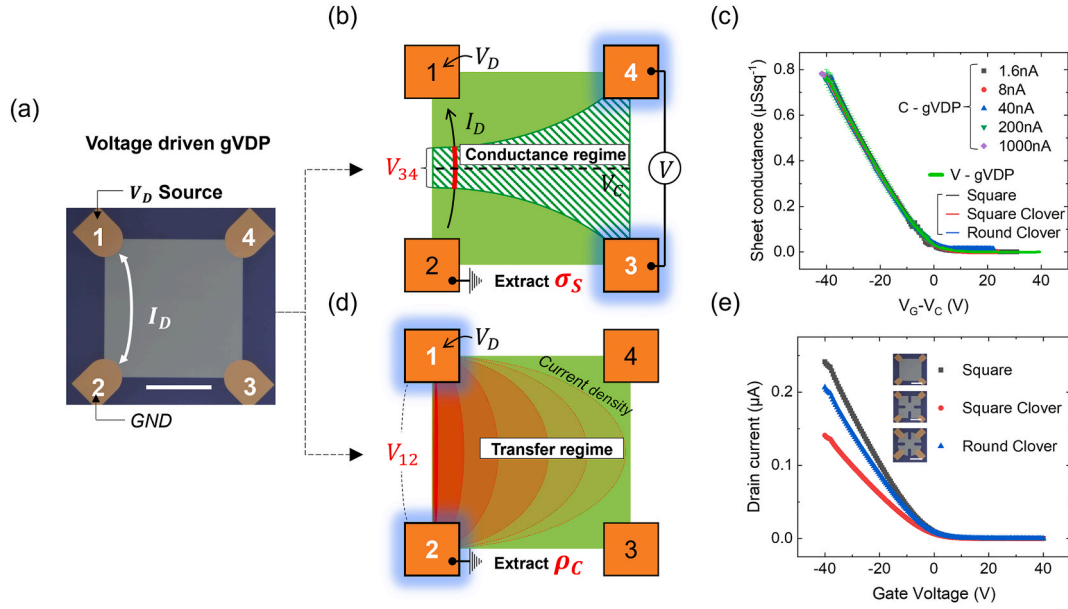


Fig. 3. (a) Top-view schematic of the V-gVDP device. The drain current (I_D) path is illustrated when a drain voltage (V_D) is applied to contact 1. The scale bar represents 0.5 mm. (b) Schematic representation of the conductance regime, where the potential difference (V_{34}) measured between contacts 3 and 4 is used to extract the sheet conductance (σ_S). No current directly flows through these sensing contacts. (c) Comparison of σ_S values for p-type SnO extracted using the C-gVDP method (at $I = 1.6$ nA, 8 nA, 40 nA, 200 nA, and 1000 nA) and the V-gVDP method ($V_D = -1$ V). The V-gVDP method is applied to gVDP devices with three geometries: Square (S.), Square Clover (S.C.), and Round Clover (R.C.). (d) Illustration of the transfer regime, where I_D measured between contacts 1 and 2 is used to extract the transfer characteristics. (e) Transfer curves measured from gVDP devices with different geometries, S., S.C., and R.C., showing that the I_D magnitude varies with geometry under identical V_G conditions.

V_C are presented in Fig. 3c for V-gVDP devices with various geometries: square (S.), square clover (S.C.), and round clover (R.C.). As V_G increases, σ_S is continuously and consistently extracted across all device shapes. Furthermore, the values closely match those obtained from the C-gVDP method, confirming that the V-gVDP approach enables reliable and geometry-independent extraction of contact-free σ_S values. A similar analysis was performed on n-type IGZO devices, as shown in Fig. S2, where the V-gVDP method produced consistent σ_S results. Fig. S3 further demonstrates that σ_S remains constant across different V_D conditions. In contrast, Fig. 3d depicts the transfer regime, where current I_{12} flows between contacts 1 and 2 through both the channel and contact electrodes, following a semi-elliptical path. The measured voltage drop (V_{12}) reflects the total device resistance, R_T , calculated as $R_T = V_{12}/I_{12}$. This resistance is observed in the transfer characteristics of the device. Although the same current I_{12} flows through both regimes, the corresponding voltage drops, V_{34} and V_{12} , capture fundamentally different physical properties: intrinsic channel resistance and total resistance (including contact), respectively. For example, while Fig. 3c confirms that σ_S remains consistent regardless of device geometry, Fig. 3e shows that R_T varies with the gVDP structure, leading to distinct transfer curves. This contrast highlights the physical separation of information obtained from each regime and reinforces the method's ability to differentiate between channel and contact properties.

The V-gVDP method thus enables the extraction of two physically distinct voltage drops, V_{34} and V_{12} , using the same drain current, allowing independent characterization of channel conductance and contact resistance. Unlike TLM, this method provides a direct, quantitative determination of σ_S and R_T from a single device, eliminating the influence of device-to-device variations. Notably, by combining these two values, the specific contact resistivity can also be accurately derived. Therefore, the V-gVDP method offers substantial advantages over both TLM and C-gVDP in terms of practicality, measurement efficiency, and analytical precision.

2.4. TCAD simulation for V-gVDPs

The conductance regime of a V-gVDP device conforms to van der Pauw's law and therefore possesses a fixed geometric ratio of $L/W = \ln 2/\pi$. In contrast, the transfer regime exhibits a semi-elliptical current path, making it theoretically difficult to define a clear L/W ratio. However, to quantitatively extract both the sheet conductance and the specific contact resistivity using the V-gVDP method, an effective L/W ratio must also be established for the transfer regime. To address this, we conducted TCAD simulations to quantitatively analyze the current flow characteristics and geometric dependencies, thereby elucidating how the effective L/W is defined in each regime. The simulations were based on square-shaped gVDP devices, and the TCAD parameters for SnO are summarized in Table 1 [41]. To examine the effect of electrode size, the electrode area was varied to 0.06 %, 0.25 %, and 1 % of the total channel area, while a drain voltage of $V_D = -1$ V was applied to contact 1. Fig. 4a–c show the resulting potential distributions for each electrode size. As the electrode area increases, the potential spread near contacts 1 and 2 becomes broader, indicating that the size of the current injection area influences the local electric field distribution. Despite these variations, the voltage difference V_{34} , measured between contacts 3 and 4 along the equipotential line, remains constant across different electrode

Table 1
Parameters used in TCAD simulation for p-type SnO TFTs.

Parameter	Value	Unit
Bandgap (E_g)	1	eV
Hole mobility (μ_p)	1	$\text{cm}^2/\text{V}\cdot\text{s}$
Relative permittivity (ϵ_r)	15	
Acceptor-like tail density (N_{TA})	2.4×10^{20}	$\text{cm}^{-3}\cdot\text{eV}^{-1}$
Donor-like tail density (N_{TD})	5.0×10^{20}	$\text{cm}^{-3}\cdot\text{eV}^{-1}$
Tail state width (Donor) (W_{TD})	0.02	eV
Tail state width (Acceptor) (W_{TA})	0.03	eV
Gaussian donor state density (N_{GD})	1×10^{17}	cm^{-3}
Gaussian donor peak energy (E_{GD})	0.9	eV
Gaussian donor width (W_{GD})	0.2	eV

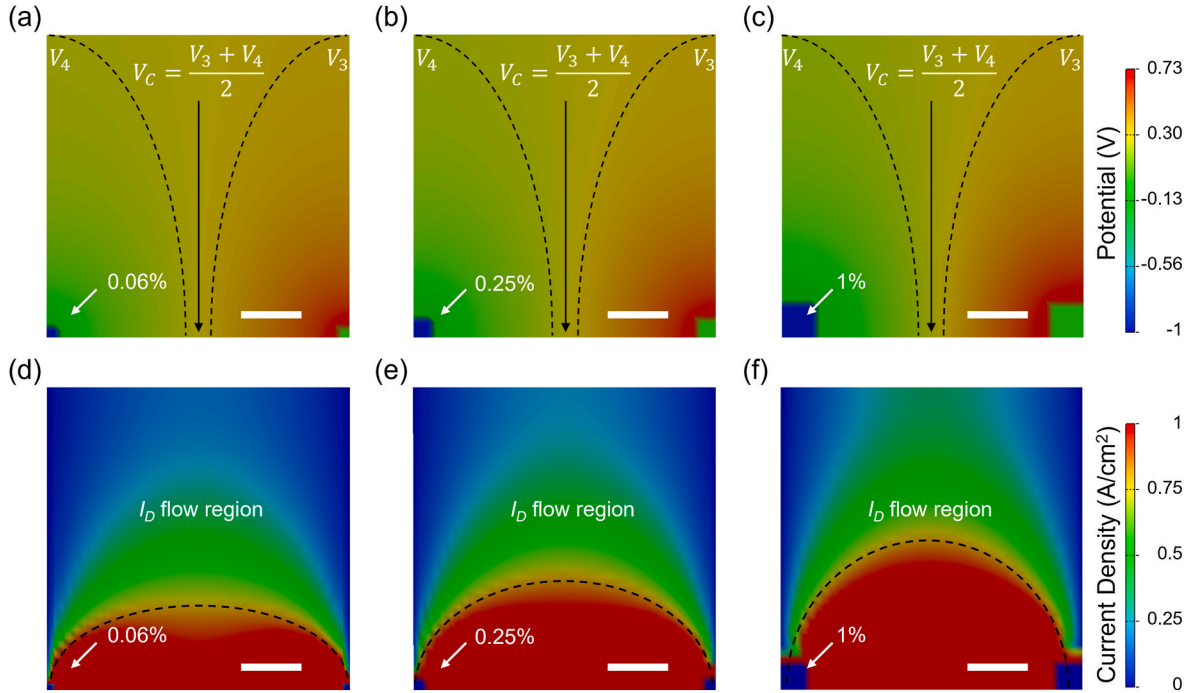


Fig. 4. TCAD simulation results for p-type SnO using gVDP devices operated under V-gVDP conditions. For gVDP structures with electrode-to-channel area ratios of 0.06 %, 0.25 %, and 1 %, (a–c) potential maps highlighting the conduction regime, while (d–f) the corresponding total current density maps, representing the transfer regime. All conditions are applied under a drain voltage of -1 V. The scale bar denotes $2 \mu\text{m}$.

sizes due to the invariant channel properties. This confirms that the conductance regime remains governed by van der Pauw's law, with a fixed $L/W = \ln 2/\pi$.

Fig. 4d–f show the corresponding current density maps. As the electrode area increases, current becomes increasingly concentrated near contacts 1 and 2, and the semi-elliptical current distribution broadens toward contacts 3 and 4. This behavior demonstrates that the current flow in the transfer regime is strongly influenced by structural parameters, such as electrode size and channel geometry. Consequently, unlike the conductance regime, the transfer regime cannot be accurately described using van der Pauw's formalism. Instead, a geometry-dependent effective L/W ratio must be defined to account for these variations. To verify the generality of this observation, TCAD simulations were also conducted for n-type IGZO under the same conditions, as presented in Fig. S4. The material parameters used are listed in Table S1 [42]. The results similarly show that while the potential distribution in the conductance regime remains stable, the current density in the transfer regime exhibits geometry-dependent variations analogous to those observed in SnO-based devices.

2.5. R_C extraction using the self-extracted L/W_{self} in V-gVDP method

To decouple R_C from R_T , we define a self-extracted effective channel geometry ratio (L/W_{self}) in the transfer regime using R_T and R_S values obtained from the V-gVDP method. Since R_{Ch} can be expressed as a product of R_S and L/W , R_T becomes:

$$R_T = R_S \frac{L}{W} + R_C \quad (1)$$

Here, R_T , R_S , and R_C are all functions of V_G . At different V_{G1} and V_{G2} , the corresponding total resistances are [43]:

$$R_T(V_{G1}) = R_S(V_{G1}) \frac{L}{W} + R_C(V_{G1}) \quad (2)$$

$$R_T(V_{G2}) = R_S(V_{G2}) \frac{L}{W} + R_C(V_{G2}) \quad (3)$$

Subtracting Eq. (3) from Eq. (2) and rearranging yields:

$$\frac{L}{W} = \frac{[R_T(V_{G1}) - R_T(V_{G2})] + [R_C(V_{G1}) - R_C(V_{G2})]}{R_S(V_{G1}) - R_S(V_{G2})} \quad (4)$$

Assuming the change in R_C is negligible ($\Delta R_C \ll \Delta R_T$, ΔR_S), this simplifies to:

$$\frac{L}{W_{\text{self}}} \approx \frac{[R_T(V_{G1}) - R_T(V_{G2})]}{R_S(V_{G1}) - R_S(V_{G2})} \quad (5)$$

Within the V-gVDP framework, R_T and R_S are extracted from the conductance and transfer regimes, respectively. These values can be combined to calculate L/W_{self} directly from the same device. Fig. 5a and d display the extracted L/W_{self} values for round clover-shaped gVDP devices based on SnO and IGZO. As the channel begins to form, L/W_{self} rapidly converges to a constant value, indicating that current flow in the semi-elliptical transfer regime stabilizes to a well-defined geometrical factor. This trend is further corroborated in Figs. S5 and S6, where both simulated and experimental devices designed with $L/W = 1$ show convergence toward the L/W_{self} values obtained from gVDP devices. Fig. 5b and e compare R_S extracted using the gVDP and TLM methods. Relative standard deviation (RSD) is provided to assess extraction precision. For p-type SnO, the RSD obtained using gVDP is 0.89 %, compared to 5.29 % from TLM, indicating a ~ 6 -fold improvement in accuracy. Similarly, for n-type IGZO, the gVDP method yields an RSD of 0.23 %, in contrast to 10.62 % for TLM, reflecting a ~ 46 -fold enhancement in precision. Notably, TLM exhibits a significantly higher RSD during the early stages of channel formation, which gradually decreases as the conduction channel becomes fully developed. This trend highlights TLM's strong reliance on linear fitting across multiple devices, a limitation that becomes apparent when the channel is not yet fully formed. In contrast, the gVDP method maintains a consistently low RSD, demonstrating greater robustness and accuracy. Comparison of the RSD values between C-gVDP and V-gVDP is presented in Fig. S7. Fig. 5c and f exhibit the quantitative decoupling of R_C from R_T by subtracting the calculated channel resistance ($R_{Ch} = R_S \cdot L/W_{\text{self}}$) from R_T . At $V_G - V_C = -30$ V, the extracted R_C for SnO is approximately 15 k Ω , while for

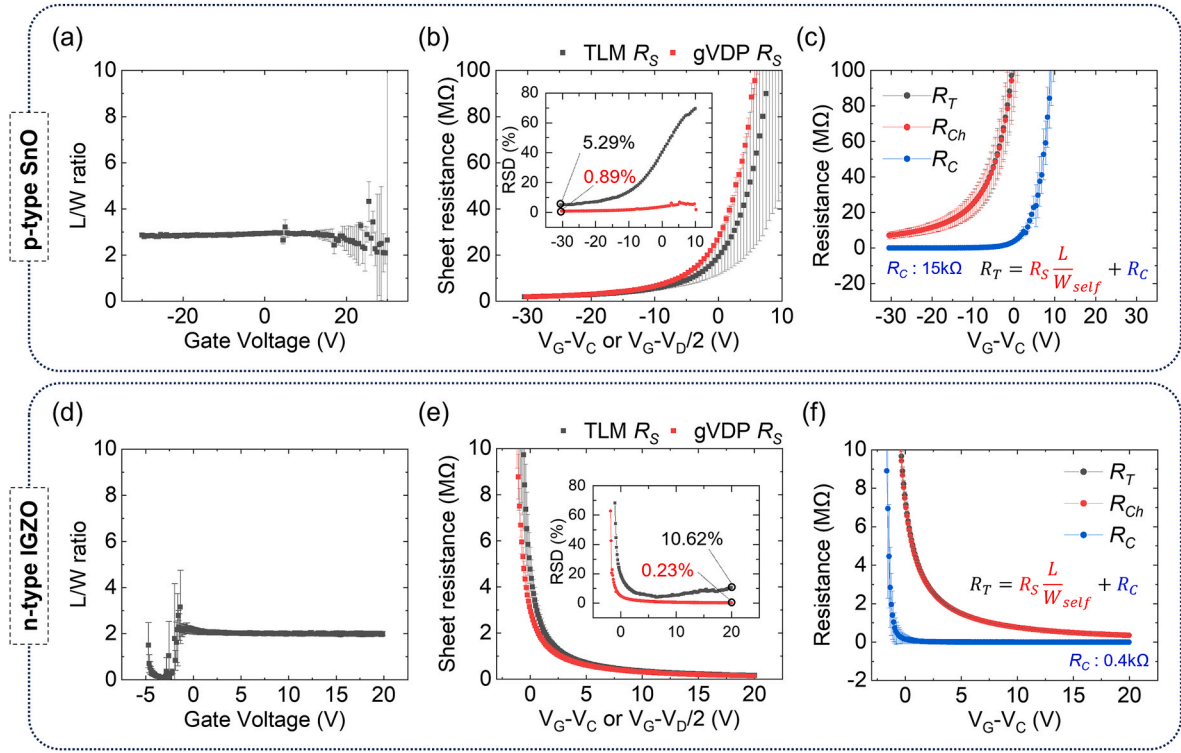


Fig. 5. Self-extracted L/W_{self} and R_C extraction using the V-gVDP. (a, d) L/W_{self} values for p-type SnO and n-type IGZO gVDP devices, respectively, showing convergence to a constant value as the channel forms. (b, e) Comparison of R_S values extracted from the V-gVDP method and the conventional TLM method for SnO and IGZO, respectively. (c, f) Decomposition of R_T into R_{Ch} and R_C . The extracted R_C values are approximately 15 kΩ for SnO at $V_G-V_C = -30$ V, and 0.4 kΩ for IGZO at $V_G-V_C = 20$ V.

IGZO at $V_G-V_C = 20$ V, it is around 0.4 kΩ. Despite both devices using the same Mo electrode ($\Phi_{Mo} \approx 4.6$ eV), the significantly higher R_C in SnO is attributed to the formation of a Schottky barrier at the contact interface. This difference arises from the higher work function of SnO, estimated to exceed 5.0 eV, compared to the work function of IGZO, approximately 4.5 eV [44–46].

These findings demonstrate that the V-gVDP method not only offers more precise extraction of R_S than TLM but also enables successful decoupling and quantification of R_C through the self-extracted L/W_{self} . This highlights the V-gVDP method as a robust, high-precision analytical platform for simultaneously characterizing channel and contact properties from a single device.

2.6. Precise specific contact resistivity (ρ_C) analysis using V-gVDP

The R_C extracted in the transfer regime using the V-gVDP method varies depending on the contact area between the electrode and the channel. Therefore, it is necessary to normalize R_C to the specific contact resistivity (ρ_C) for quantitative comparison across devices. In the TLM method, ρ_C is defined as $\rho_C = \frac{(R_C W)^2}{R_S}$, where W is considered the effective current injection width. In contrast, for the gVDP structure, the effective current injection width is defined as the perimeter of the electrode/channel interface. Fig. 6a shows the ρ_C extracted by the V-gVDP method for the p-type SnO device, which is approximately $1.1 \times 10^{-1} \Omega \text{ cm}^2$ at V_G-V_C or $V_G-V_D/2 = -30$ V, with a RSD of $\sim 14\%$. On the other hand, the TLM method defines R_C by linear extrapolation at $L = 0$, resulting in negative R_C values in the -30 V to -27 V range, making ρ_C calculation infeasible in this region. Even within the valid range of -26 V to -15 V,

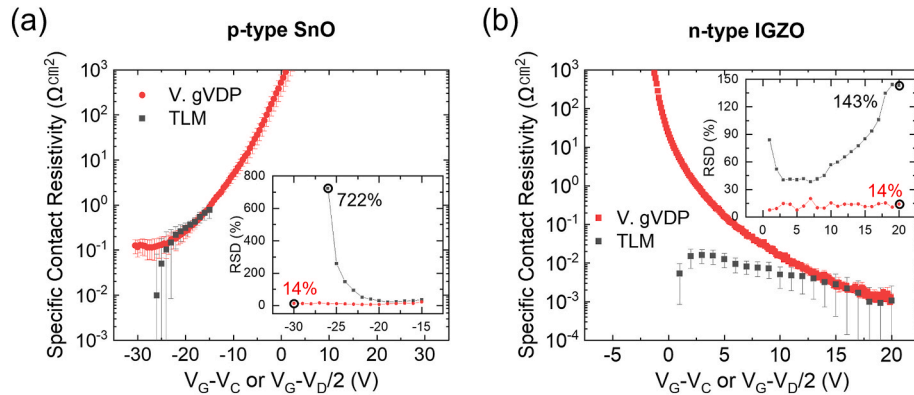


Fig. 6. Comparison of ρ_C extracted by V-gVDP and TLM for (a) p-type SnO and (b) n-type IGZO gVDP devices. The V-gVDP method yields a ρ_C of approximately $1.1 \times 10^{-1} \Omega \text{ cm}^2$ for SnO and $1.3 \times 10^{-3} \Omega \text{ cm}^2$ for IGZO. Both devices utilize Mo electrodes with a work function of 4.6 eV.

the extracted ρ_C aligns with the V-gVDP values up to -22 V but rapidly decreases as $V_G - V_C$ or $V_G - V_D/2$ approaches -26 V, ultimately yielding an unphysically low value of approximately $9.86 \times 10^{-3} \Omega \text{ cm}^2$. At this point, the RSD reaches 722 %, approximately 51 times higher than that of V-gVDP, indicating a significant loss of accuracy. This suggests that the TLM method, which relies on multi-device fitting, is particularly vulnerable when analyzing unstable devices like SnO. In contrast, the V-gVDP method enables the continuous extraction of ρ_C from the point at which the channel begins to form using a single device, independent of device-to-device trends. Although switching the current direction ($I_{12} \rightarrow I_{21}$) may introduce a slight V_T shift, the device characteristics are extracted with consistent trends, maintaining an RSD of approximately 14 %. Fig. 6b presents the results for the n-type IGZO device. The ρ_C extracted using the V-gVDP method is $\sim 1.3 \times 10^{-3} \Omega \text{ cm}^2$ at $V_G - V_C$ or $V_G - V_D/2 = 20$ V, with an RSD of 14 %, while the TLM method yields a similar value of $\sim 1.0 \times 10^{-3} \Omega \text{ cm}^2$ but with a much larger RSD of 143 %. Notably, below $V_G - V_C$ or $V_G - V_D/2 \approx 11$ V, the two methods exhibit divergent trends. As confirmed in Fig. S7, the TLM fitting lines no longer converge to a single point in this low- V_G region. This is attributed to the failure to account for V_T shifts caused by different channel lengths. As a result, the R_S extracted by fitting does not match the measured wR_T of each device. Furthermore, as TLM includes devices with large parameter variations, the reliability of the extracted values degrades significantly. In contrast, the V-gVDP method, as demonstrated for SnO, also allows the IGZO device to extract reliable and continuous ρ_C from the onset of channel formation, with an RSD maintained at ~ 14 %. The RSD values for TLM, C-gVDP, and V-gVDP are summarized in Table 2. These results confirm that the V-gVDP method is a robust and practical analysis technique for precise extraction of both σ_S and ρ_C . In particular, the consistent accuracy achieved in both p-type SnO and n-type IGZO TFTs highlights the broad applicability of this method across a range of oxide semiconductors and its utility as a valuable tool for designing next-generation oxide CMOS technologies.

3. Conclusion

In this study, we propose the V-gVDP method, which is capable of simultaneously and precisely extracting both channel and contact properties from a single device, without the need for additional control devices. We validated its effectiveness by applying it to p-type SnO and n-type IGZO oxide TFTs. Unlike the conventional TLM, which is highly sensitive to device-to-device variations and particularly unreliable for devices with high R_C , such as p-type TFTs, the V-gVDP method fundamentally overcomes these limitations by spatially separating and independently analyzing the conductance and transfer regimes within a single device. Quantitatively, the RSD value of the sheet resistance extracted using the V-gVDP method was approximately 6 times lower for SnO and 46 times lower for IGZO compared to TLM. For the specific contact resistivity, the V-gVDP achieved a 51-fold improvement in precision for SnO and a 10-fold improvement for IGZO. Moreover, the V-gVDP method produced no physically invalid results, such as the negative R_C values frequently observed in TLM analyses. In particular, while the TLM method failed to extract ρ_C in the range of -30 V to -27 V for SnO devices, the V-gVDP method enabled continuous extraction from the onset of channel formation, maintaining a stable RSD of approximately 14 % throughout. Furthermore, the V-gVDP approach

offers a significantly simplified measurement setup compared to the current-driven C-gVDP method, as it eliminates the need for complex current segmentation or separate reference devices for R_C extraction. By introducing a self-extracted effective geometry factor (L/W_{self}), the V-gVDP method enables quantitative decoupling of R_C from R_T , thereby facilitating high-precision characterization of both intrinsic channel and contact properties. In conclusion, the V-gVDP method provides a robust and accurate analytical platform that effectively eliminates the influence of device variability and allows reliable extraction of both σ_S and ρ_C from a single device. Its general applicability to a wide range of oxide semiconductors makes it a highly practical tool for advanced oxide CMOS circuit design, contact engineering optimization, and the development of low-power, high-reliability electronic devices.

4. Experimental section/methods

4.1. Fabrication

All devices were fabricated on heavily doped p-type Si wafers with a 100 nm thermally grown SiO_2 layer used as the gate dielectric, forming a conventional Si gate structure. Before fabrication, SC-1 and SC-2 cleaning processes were performed. A 20 nm-thick SnO film was deposited via reactive RF sputtering (located at the 3D Convergence Center of Inha University) using a metallic Sn target in an Ar/ O_2 ambient with 10 % oxygen partial pressure. The sputtering was carried out at a working pressure of 3 mTorr and an RF power of 120 W. The SnO film was patterned using a lift-off process and subsequently annealed in air at 225°C for 1 h to improve crystallinity and stabilize the electrical properties. The a-IGZO film was also deposited by RF sputtering under an Ar flow of 20 sccm, followed by annealing in air at 350°C for 1 h in a furnace. Patterning was performed by wet etching using an HCl solution. Molybdenum (Mo) was used as the source/drain metal and deposited by DC sputtering, followed by patterning via a lift-off process. All fabrication steps were performed simultaneously for both the TLM and gVDP devices on the same substrate, and both devices shared the same Si common gate.

4.2. Characterization

All devices were characterized in ambient air using a Keithley 4200 parameter analyzer connected to a probe station. TLM devices were measured under fixed ground/voltage/voltage sweep conditions applied to the source/drain/gate electrodes, respectively, depending on the channel length. The C-gVDP devices were measured using a van der Pauw-based configuration. Current was applied to contact 1, contact 2 was grounded, and contacts 3 and 4 were used to measure the gate voltage. Positive and negative current injections were applied for the SnO and a-IGZO devices, respectively, considering their p-type and n-type characteristics.

The injection currents were set to 1.6 nA, 8 nA, 40 nA, 200 nA, and 1000 nA. To reduce variability caused by defects, eight measurements were performed for each current level by alternating the roles of the four contacts. The sheet conductance at each current was then aggregated to extract the saturated sheet conductance as a function of gate voltage. The V-gVDP devices were measured in a manner similar to the C-gVDP method, but with voltage applied instead of current. Transfer curves

Table 2
Comparison of R_S , ρ_C , and RSD extracted by TLM, C-gVDP, and V-gVDP methods.

SnO @ -30 V		TLM		C-gVDP		V-gVDP	
IGZO @ 20 V		Value	RSD	Value	RSD	Value	RSD
Sheet Resistance (R_S)	SnO	$1.91 \pm 0.10 \text{ M}\Omega$	5.29 %	$1.77 \pm 0.03 \text{ M}\Omega$	1.46 %	$1.87 \pm 0.02 \text{ M}\Omega$	0.89 %
	IGZO	$0.17 \pm 0.10 \text{ M}\Omega$	10.62 %	$0.16 \pm 0.00 \text{ M}\Omega$	2.66 %	$0.15 \pm 0.00 \text{ M}\Omega$	0.23 %
Specific Contact Resistivity (ρ_C)	SnO	$9.9\text{E-}3 \pm 0.7\text{E-}1 \Omega \text{ cm}^2$	722 %	–	–	$1.2\text{E-}1 \pm 4.4\text{E-}2 \Omega \text{ cm}^2$	14 %
	IGZO	$1.1\text{E-}3 \pm 1.5\text{E-}3 \Omega \text{ cm}^2$	143 %	–	–	$1.3\text{E-}3 \pm 3.3\text{E-}4 \Omega \text{ cm}^2$	14 %

were obtained from contacts 1 and 2, while sheet conductance curves were extracted from contacts 3 and 4.

The precision and reproducibility of the voltage-driven gVDP method were evaluated by comparing sheet conductance extracted from gVDP devices with square, square-clover, and round-clover geometries, and calculating the relative standard deviation ($RSD = (\text{standard deviation}/\text{mean value}) \times 100\%$) according to device geometry.

Five devices were fabricated for each of several different channel lengths, and their electrical performance was evaluated to statistically calculate the RSD values. For the SnO-TFTs, four different channel lengths were studied using five devices per length, resulting in a total of 20 samples. In the case of IGZO-TFTs, six channel lengths were investigated under the same conditions, yielding 30 samples in total. The difference in channel length selection between SnO and IGZO devices arises from their relative reliability. SnO-TFTs exhibited unstable behavior at longer channel lengths. Therefore, TLM structures with channel lengths ranging from 10 to 75 μm were employed to ensure consistent and reliable measurements. In contrast, IGZO-TFTs demonstrated stable and reproducible electrical characteristics across a broader range, allowing for reliable evaluation over channel lengths from 10 to 120 μm .

CRediT authorship contribution statement

Woo-Seok Lee: Writing – original draft, Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **Arqum Ali:** Methodology, Investigation. **Jaeho Lee:** Visualization, Investigation. **Jiyoung Kim:** Writing – review & editing, Data curation. **Rino Choi:** Writing – review & editing, Supervision, Funding acquisition. **Jeong-Hwan Lee:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.mtphys.2025.101880>.

Data availability

Data will be made available on request.

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